

BahaAI — Manual Test Walkthrough

Every function, from a clean install to adviser sign-off

An IoT-Based Flood Alert System · Municipality of Consolacion, Cebu
University of Cebu Lapu-Lapu and Mandaue · Prepared 20 September 2026

HOW TO USE THIS DOCUMENT

Work through the parts **in order** — later tests depend on data created by earlier ones. Each test gives the role required, the exact steps, and the exact result to expect. Tick **Pass** or **Fail** and write a note for anything that fails.

Allow about **90 minutes** for a complete pass. Part 0 must be finished before anything else.

BEFORE YOU DEMONSTRATE THIS TO YOUR ADVISER

Run the whole walkthrough yourself first and fix anything that fails. A test marked Pass that fails in front of a panel costs more than an honest Fail written down in advance.

Role legend

Each test shows the account you must be signed in as:

ADMIN Administrator **STAFF** DRRMO / Staff **RESIDENT** Community Resident **GUEST** Not signed in **TERMINAL** Command line

Test accounts

Role	Email	Password
Administrator	admin@bahaai.test	password
DRRMO / Staff	See Admin → Users , filter by Staff	password
Resident	See Admin → Users , filter by Resident	password

0 Setting up from nothing

Nothing but Docker Desktop is required. No PHP, Node, Composer or PostgreSQL on the machine.

T0.1 Install Docker Desktop

SETUP

1. Download Docker Desktop from docker.com/products/docker-desktop.
2. **Windows:** install, then restart the computer. Allow WSL 2 if prompted.
3. Open Docker Desktop and wait for the bottom-left indicator.

Expected: Docker Desktop shows **Engine running**.

☐ Pass ☐ Fail Notes: _____

T0.2 Start the application

TERMINAL

1. Open a terminal in the project folder (the one containing `docker-compose.yml`).
2. Run a clean first build:

```
docker compose up --build
```

Expected: after a few minutes the log ends with `BahaAI is running -> http://localhost:8000`. Migrations run automatically and demo data is seeded on the first start only.

☐ Pass ☐ Fail Notes: _____

T0.3 Confirm the seeded data

TERMINAL

1. In a second terminal, run:

```
docker compose exec app php artisan tinker --execute='
foreach
(["users", "flood_zones", "sensors", "sensor_readings", "actuator_devices", "evacuation_centers", "flood_incidents"] as
$t) {
    printf("%-20s %d\n", $t, DB::table($t)->count());
}'
```

Expected: 24 users, 7 flood zones, 13 sensors, thousands of readings, 7 actuator devices, **8 evacuation centers**, 25 flood incidents. Zero of anything listed means the seed did not run.

☐ Pass ☐ Fail Notes: _____

T0.4 Open the application and sign in

GUEST

1. Browse to `http://localhost:8000`.
2. Sign in as `admin@bahaai.test` / `password`.

Expected: the Administrator dashboard loads with live sensor counts and a sidebar containing Dashboard through GPS Flood Alerts, plus an admin group with Flood Zones, Sensors, Evac. Centers, Users and Audit Log.

☐ Pass ☐ Fail Notes: _____

T0.5 Open the mail catcher

SETUP

1. Open `http://localhost:8025` in a second browser tab and keep it open.

Expected: the Mailpit inbox loads. Every email the system sends appears here; nothing reaches a real inbox.

☐ Pass ☐ Fail Notes: _____

1 Account Management

Module 1 — registration with one-time-password verification, sign-in, profile and location.

T1.1 Register a new resident account

GUEST

1. Sign out. From the login page choose **Register here**.
2. Enter a name, an email you have not used, a mobile number and a password.
3. Submit.

Expected: you are taken to the **Verify OTP** screen, not straight into the app.

☐ Pass ☐ Fail Notes: _____

T1.2 Receive and use the one-time password

GUEST

1. Switch to the Mailpit tab at `http://localhost:8025`.
2. Open the newest message and note the 6-digit code.
3. Enter it on the Verify OTP screen.

Expected: a branded email containing a 6-digit code arrives within seconds. Entering it signs you in and lands you on the resident dashboard.

☐ Pass ☐ Fail Notes: _____

T1.3 A wrong code is rejected

GUEST

1. Register a second account.
2. On the Verify OTP screen enter `000000`.

Expected: a validation error appears and you stay on the OTP screen. You are not signed in.

☐ Pass ☐ Fail Notes: _____

T1.4 Resend is rate limited

GUEST

1. On the Verify OTP screen press **Resend** four times in quick succession.

Expected: the first three succeed; the fourth is refused with a "too many attempts" message. This proves the throttle protects against email flooding.

☐ Pass ☐ Fail Notes: _____

T1.5 Each role lands on the right dashboard

GUEST

1. Sign in as the administrator, note the sidebar, sign out.
2. Repeat as a staff account, then as a resident account.

Expected: Admin sees every item including the admin group. Staff sees operational items (Alerts, Analytics, Smart Actuation, Flood Reports) but no admin group. Resident sees monitoring and information items only — no Analytics, no Smart Actuation, no admin group.

☐ Pass ☐ Fail Notes: _____

T1.6 Password reset

GUEST

1. From the login page choose **Forgot Password?** and submit a known email.
2. Open the reset link from Mailpit and set a new password.
3. Sign in with the new password.

Expected: the reset email arrives, the link opens a reset form, and the new password works while the old one no longer does.

☐ Pass ☐ Fail Notes: _____

T1.7 Update the profile

RESIDENT

1. Open **Profile & Settings** from the sidebar.
2. Change the name and mobile number, set the barangay to **Tayud**, and save.

Expected: a success message appears and the values persist after a page reload. Setting the barangay matters — it decides which alerts this account receives.

☐ Pass ☐ Fail Notes: _____

T1.8 Notification preferences

RESIDENT

1. On the profile page find **Notification Preferences**.
2. Enable SMS and email, disable push, and save.

Expected: the choices persist after reload. These decide which channel rows appear in the notification log in Part 7.

☐ Pass ☐ Fail Notes: _____

T1.9 Set the monitored location

RESIDENT

1. Choose **Change Location** at the bottom of the sidebar.
2. Set a latitude and longitude inside Consolacion, for example `10.3600` and `123.9500`, and save.

Expected: the location saves. This is what lets the Evacuation Centers page sort shelters nearest-first in Part 11.

☐ Pass ☐ Fail Notes: _____

2 Dashboard and Visualisation

Module 2 — live dashboard, water levels, GIS map and predictive forecasts.

T2.1 Dashboard shows live figures

STAFF

1. Open **Dashboard**.

Expected: sensor counts, a breakdown of zones by risk level, recent alerts and a water level chart. Figures must be non-zero — they come from the seeded readings.

☐ Pass ☐ Fail Notes: _____

T2.2 Water levels list and detail

RESIDENT

1. Open **Real-time Monitoring**.
2. Open any sensor to see its detail page.
3. Change the time range (for example 24 hours to 7 days).

Expected: each sensor shows its latest level with a safe, warning or critical badge. The detail page shows a time-series chart that redraws when the range changes.

☐ Pass ☐ Fail Notes: _____

T2.3 GIS flood map

RESIDENT

1. Open **Flood Map**.
2. Zoom and pan. Click a zone polygon, then a sensor marker.

Expected: a real street map of Consolacion with zone polygons coloured by risk (green, amber, red) and sensor markers. Clicking either opens a popup with its details.

☐ Pass ☐ Fail Notes: _____

T2.4 Weather

RESIDENT

1. Open **Rainfall & Weather**.

Expected: current conditions for Consolacion with temperature, humidity, rainfall and wind. With no API key configured these values are simulated — say so if asked.

☐ Pass ☐ Fail Notes: _____

3 Notifications and Alerts interface

Module 3 — the resident-facing inbox.

T3.1 Notification inbox and unread badge

RESIDENT

1. Look at the bell icon in the top bar and note the unread count.
2. Open **Notifications**.
3. Click an unread item.

Expected: unread entries are visually distinct. Opening one marks it read and the bell count decreases by one.

☐ Pass ☐ Fail Notes: _____

1. On the Notifications page choose **Mark all as read**.

Expected: every entry becomes read and the bell badge disappears.

☐ Pass ☐ Fail Notes: _____

4 IoT Telemetry and Device Management

Module 4 — sensor records, device authentication, battery reporting and reachability. This part proves the path a real ESP32 will use.

T4.1 Create, edit and delete a sensor

ADMIN

1. Open **Sensors** in the admin group and choose **Add Sensor**.
2. Create one named `Test Sensor A` in any zone, then edit its name and save.
3. Delete it.

Expected: each operation shows a success message and the list updates. Submitting the form with a blank name shows a validation error instead of saving.

☐ Pass ☐ Fail Notes: _____

T4.2 Issue a device token

TERMINAL

1. Mint a token for sensor 1 and copy the printed value:

```
docker compose exec app php artisan sensors:token 1
```

Expected: a 48-character token is printed once, with a warning that it is stored hashed and cannot be shown again.

☐ Pass ☐ Fail Notes: _____

T4.3 A reading with no token is rejected

TERMINAL

1. Post a reading without an `Authorization` header:

```
curl.exe -i -X POST http://localhost:8000/api/sensors/1/readings ^  
-H "Content-Type: application/json" -H "Accept: application/json" ^  
-d "{\"water_level\":9.9}"
```

Expected: HTTP 401. No reading is stored and no alert is raised. This is the security fix: without it, anyone could fabricate a flood and trigger real evacuation messages.

☐ Pass ☐ Fail Notes: _____

T4.4 A reading with a valid token is accepted

TERMINAL

1. Repeat with the token from T4.2 and include a battery level:

```
curl.exe -i -X POST http://localhost:8000/api/sensors/1/readings ^  
-H "Content-Type: application/json" -H "Accept: application/json" ^  
-H "Authorization: Bearer PASTE_TOKEN_HERE" ^  
-d "{\"water_level\":1.42,\"temperature\":28.4,\"humidity\":81,\"battery_level\":76.5}"
```

Expected: HTTP 201 with the created reading in the response body.

☐ Pass ☐ Fail Notes: _____

POWERSHELL NOTE

In PowerShell, `curl` is an alias for a different command. Type `curl.exe` with the extension, and use `^` for line continuation as shown. In Git Bash use `curl` and `\` instead.

T4.5 Battery level appears in the sensor list

ADMIN

1. Open **Sensors** and find the row for sensor 1.

Expected: the Battery column shows **77%** as a green badge. Sensors that have never reported show "not reported" rather than a fabricated figure.

☐ Pass ☐ Fail Notes: _____

T4.6 An impossible battery value is refused

TERMINAL

1. Repeat T4.4 with `"battery_level":150`.

Expected: **HTTP 422** with the message "The battery level field must be between 0 and 100."

☐ Pass ☐ Fail Notes: _____

T4.7 A silent sensor is marked unreachable

TERMINAL

1. Age a sensor artificially, then run the offline check:

```
docker compose exec app php artisan tinker --execute='
App\Models\Sensor::find(2)->update(["last_reading_at" => now()->subHours(3)]);'
docker compose exec app php artisan sensors:check-offline
```

Expected: the command reports a new offline alert. In **Admin** → **Sensors** that sensor now shows status **offline** in red, and an alert appears on the Alerts page.

☐ Pass ☐ Fail Notes: _____

T4.8 Reporting again brings it back online

TERMINAL

1. Run the simulator, which represents every device reporting again:

```
docker compose exec app php artisan sensors:simulate
```

Expected: the sensor from T4.7 returns to status **active**.

☐ Pass ☐ Fail Notes: _____

5 Smart Actuation

Module 5 — the Smart Drainage Module. Pumps, valves and floodgates engage at critical; sirens engage at warning; everything releases when the water recedes.

T5.1 Set every device to automatic

STAFF

1. Open **Smart Actuation**.
2. Set each device's mode to **Auto**.

Expected: each change is confirmed, and the Start/Stop button becomes disabled for devices in auto mode with the note "Switch to manual mode to control directly."

☐ Pass ☐ Fail Notes: _____

T5.2 A storm engages the hardware automatically

TERMINAL

1. Drive every zone to warning or critical:

```
docker compose exec app php artisan sensors:simulate --storm
```

Expected: alerts are created. On **Smart Actuation**, pumps in critical zones now show **ON**, and sirens in zones at warning or above show **ON** — without anyone pressing a button.

☐ Pass ☐ Fail Notes: _____

T5.3 The activity log attributes it to the system

STAFF

1. Scroll to the activity log on the Smart Actuation page.

Expected: recent entries are tagged **automatic** rather than manual, with a note such as "Zone risk critical — auto engaged". Manual entries carry the name of the operator.

☐ Pass ☐ Fail Notes: _____

T5.4 Hardware releases when the water recedes

TERMINAL

1. Generate calm readings:

```
docker compose exec app php artisan sensors:simulate
```

Expected: every auto device returns to **off**, with log entries reading "Zone risk safe — auto released". Automation that only switches on is not automation.

☐ Pass ☐ Fail Notes: _____

T5.5 Manual control works in manual mode

STAFF

1. Switch one pump to **Manual**.
2. Press **Start**, then **Stop**.

Expected: the device switches on and off, and two log entries appear tagged **manual** with your name against them.

☐ Pass ☐ Fail Notes: _____

T5.6 Residents cannot reach the control panel

RESIDENT

1. Sign in as a resident.
2. Browse directly to `http://localhost:8000/actuation`.

Expected: 403 Forbidden. The item is also absent from the resident sidebar.

☐ Pass ☐ Fail Notes: _____

6 Flood Prediction

Module 6 — linear regression over recent readings producing rate of rise, time-to-critical and a confidence score.

T6.1 Generate predictions

TERMINAL

1. Create a rising trend, then generate predictions:

```
docker compose exec app php artisan sensors:simulate
docker compose exec app php artisan sensors:simulate --storm
docker compose exec app php artisan predictions:generate
```

Expected: the command reports predictions generated for the active sensors.

☐ Pass ☐ Fail Notes: _____

T6.2 Read the forecast

STAFF

1. Open **AI Flood Prediction**.

Expected: each sensor shows its current level, rate of rise in metres per minute, a projected level, a time-to-critical estimate where the water is rising, and a confidence gauge.

☐ Pass ☐ Fail Notes: _____

BE READY FOR THIS QUESTION

If an adviser asks what the confidence percentage means, the correct answer is: **it is $R^2 \times 100$ — how closely the recent readings fit a straight line, not the probability that the forecast is correct.** A steady rise gives high confidence; an erratic one gives low confidence. Saying this plainly is stronger than overclaiming.

7 Automated Alert Dispatch

Module 7 — threshold breach to alert to multi-channel dispatch, with a per-recipient delivery log.

T7.1 A breach creates an alert

TERMINAL

1. Run `docker compose exec app php artisan sensors:simulate --storm`.
2. Open **Alerts & Notifications** as staff.

Expected: new alerts appear with severity warning or critical, naming the sensor, the recorded level, the threshold breached and the barangay.

☐ Pass ☐ Fail Notes: _____

T7.2 The delivery log names every recipient and channel

STAFF

1. Open any critical alert and find the **Notification Log** panel.

Expected: one row per recipient per channel. Email rows show **sent** in green. SMS rows show **simulated** in grey. Siren rows name the device rather than a person.

☐ Pass ☐ Fail Notes: _____

THIS IS THE HONESTY POINT TO HIGHLIGHT

SMS is recorded as **simulated**, never as sent, and carries no delivery timestamp — so a simulated message can never be mistaken for a delivered one. Real SMS can be enabled by setting `SMS_GATEWAY=semaphore` with a provider key; no code changes are required.

T7.3 Residents really receive the email

GUEST

1. Switch to the Mailpit tab at `http://localhost:8025`.

Expected: a branded flood alert email for each affected resident, containing the severity, zone, water level and guidance. Email is genuinely delivered, not simulated.

☐ Pass ☐ Fail Notes: _____

T7.4 Alerts are not duplicated

TERMINAL

1. Run `--storm` twice more and count the alerts for one sensor.

Expected: no second alert of the same severity is created for the same sensor while the first is still active. This is deliberate: without it, residents would be messaged every five minutes for the whole event.

☐ Pass ☐ Fail Notes: _____

T7.5 Staff can issue a manual alert

STAFF

1. On the Alerts page choose **Create Alert**.
2. Pick a zone and severity, write a message, and submit.

Expected: the alert is created with source **manual** and your name recorded as its creator, and notifications dispatch exactly as for an automatic alert.

☐ Pass ☐ Fail Notes: _____

T7.6 Resolving an alert records a flood incident

STAFF

1. Open an active **critical** alert and choose **Resolve**.
2. Then run:

```
docker compose exec app php artisan tinker --execute='
$i = App\Models\FloodIncident::latest("id")->first();
printf("peak=%s rainfall=%s residents=%d duration=%d min\n",
    $i->peak_water_level, $i->total_rainfall, $i->affected_residents, $i->duration_minutes);'
```

Expected: a flood incident now exists whose peak level and rainfall come from the readings recorded while that alert was active, and whose affected-resident count is the real number of residents registered in that barangay.

☐ Pass ☐ Fail Notes: _____

8 Reports and Historical Analytics

Module 8 — analytics, historical review and export.

T8.1 Analytics with a date filter

STAFF

1. Open **Reports & Analytics**.
2. Set a date range covering the last 30 days and apply it.

Expected: total incidents, critical incidents, affected residents and peak level update with the range, plus a per-zone breakdown.

☐ Pass ☐ Fail Notes: _____

T8.2 Export to CSV

STAFF

1. Choose **Export** and select CSV.
2. Open the file in a spreadsheet.

Expected: a CSV downloads containing a header row and one row per incident, with zone, barangay, severity, peak level, rainfall and date.

☐ Pass ☐ Fail Notes: _____

T8.3 Export to PDF

STAFF

1. Choose **Export** and select PDF.

Expected: a formatted PDF report downloads and opens, with a title, the date range and the incident table.

☐ Pass ☐ Fail Notes: _____

T8.4 Historical data view

STAFF

1. Open **Historical Data** and change the date range.

Expected: readings and flood events aggregated over the chosen period, with charts that redraw when the range changes.

☐ Pass ☐ Fail Notes: _____

9 User and Access Management

Module 9 — accounts and roles.

T9.1 Search and filter users

ADMIN

1. Open **Users**. Search for part of a name, then filter by role.

Expected: the list narrows correctly and the filters combine.

☐ Pass ☐ Fail Notes: _____

T9.2 Create an account

ADMIN

1. Choose **Add User**, create a staff account, and sign in as it in a private window.

Expected: the account is created pre-verified — it skips the OTP step — and the staff sidebar appears. Reusing an existing email shows a validation error.

☐ Pass ☐ Fail Notes: _____

T9.3 Change an existing user's role

ADMIN

1. Choose the pencil icon beside any resident.
2. Change the role to **DRRMO / Staff**, leave the password fields blank, and save.

Expected: the role changes and the user's existing password still works. Leaving the password blank must not clear it.

☐ Pass ☐ Fail Notes: _____

T9.4 An administrator cannot demote themselves

ADMIN

1. Edit your own account and try to change the role to Resident.

Expected: refused with "You cannot change your own role." This prevents the last administrator locking everyone out.

☐ Pass ☐ Fail Notes: _____

T9.5 Role enforcement holds on direct URLs

RESIDENT

1. Signed in as a resident, visit each URL below directly.

URL	Expected
/admin/users	403 Forbidden
/admin/audit-logs	403 Forbidden
/admin/evacuation-centers	403 Forbidden
/analytics	403 Forbidden
/actuation	403 Forbidden
/dashboard	Loads normally

Expected: hiding a menu item is not security. Each restricted URL must be refused by the server even when typed directly.

☐ Pass ☐ Fail Notes: _____

10 System Configuration

Module 10 — flood zones and the thresholds that drive every alert.

T10.1 Flood zone create, edit and delete

ADMIN

1. Open **Flood Zones** and add a test zone with at least four boundary points.
2. Edit its description, then delete it.

Expected: each operation succeeds with a confirmation. The new zone appears on the Flood Map while it exists.

☐ Pass ☐ Fail Notes: _____

T10.2 Thresholds must be ordered

ADMIN

1. Edit a zone and set warning **3.0** with critical **1.5**. Save.

Expected: a validation error. Critical must exceed warning, which must exceed safe — otherwise risk classification would be nonsense.

☐ Pass ☐ Fail Notes: _____

T10.3 Thresholds change classification immediately

ADMIN

1. Note a zone currently showing **safe** and its latest water level.
2. Edit the zone and set the warning threshold below that level. Save.
3. Return to Real-time Monitoring.

Expected: the zone and its sensors now classify as **warning**, with no new reading required. Restore the original values afterwards.

☐ Pass ☐ Fail Notes: _____

11 Evacuation Center Management

Module 11 — shelter records, capacity and nearest-shelter selection.

T11.1 The shelter register

ADMIN

1. Open **Evac. Centers** in the admin group.

Expected: eight seeded shelters across the barangays, with total capacity and spaces available shown as summary figures, and an occupancy bar per shelter.

☐ Pass ☐ Fail Notes: _____

T11.2 Create, edit and delete a shelter

ADMIN

1. Choose **Add Center**, create one in Tayud with capacity 200, and save.
2. Edit its occupancy to 150, then delete it.

Expected: each operation succeeds; the occupancy bar reflects 75 per cent while it exists.

☐ Pass ☐ Fail Notes: _____

T11.3 Occupancy cannot exceed capacity

ADMIN

1. Edit a shelter and set occupancy above its capacity. Save.

Expected: "Occupancy cannot exceed the centre capacity."

☐ Pass ☐ Fail Notes: _____

T11.4 Residents see shelters nearest first

RESIDENT

1. Signed in as the resident whose location you set in T1.9, open **Evacuation Centers**.

Expected: a real street map with shelter markers coloured by availability, a **Nearest Center** card naming a shelter with its distance in kilometres and spaces free, a working **Get directions** link, and a full list ordered by distance.

☐ Pass ☐ Fail Notes: _____

T11.5 A full shelter is not offered

ADMIN

1. Set the nearest shelter from T11.4 to status **full** and save.
2. Return to the resident's Evacuation Centers page, then open **GPS Flood Alerts** as staff.

Expected: the full shelter is marked accordingly, and the evacuation route now targets the nearest shelter that is still **open** rather than the closest one overall.

☐ Pass ☐ Fail Notes: _____

12 Audit and Compliance

Module 12 — who changed what, when, and from where.

T12.1 Changes are recorded with the actor

ADMIN

1. Open **Audit Log** and note the current count.
2. Edit any flood zone's critical threshold and save.
3. Return to the Audit Log.

Expected: a new entry showing the time, your name, action **updated**, the record type and name, and the originating IP address.

☐ Pass ☐ Fail Notes: _____

T12.2 Before and after values are captured

ADMIN

1. Click that entry to expand it.

Expected: a field-by-field table showing the old value in red and the new value in green — for example `critical_threshold 3.00 → 4.50`.

☐ Pass ☐ Fail Notes: _____

T12.3 Creations and deletions are recorded too

ADMIN

1. Create a shelter, then delete it. Return to the Audit Log.

Expected: one **created** entry and one **deleted** entry, both naming the shelter. Deleted records remain traceable after the record itself is gone.

☐ Pass ☐ Fail Notes: _____

T12.4 Credentials never appear in the trail

ADMIN

1. Edit a user and set a new password. Save.
2. Expand the resulting audit entry.

Expected: the password field shows `[redacted]`. Device API tokens are redacted the same way. An audit trail that leaks credentials is worse than none.

☐ Pass ☐ Fail Notes: _____

T12.5 Filters work

ADMIN

1. Filter by action **updated**, then by record type, then search a record name.

Expected: the list narrows correctly and the filters combine.

☐ Pass ☐ Fail Notes: _____

13 Community Flood Reports

Residents report what they see; staff verify it.

T13.1 A resident submits a report**RESIDENT**

1. On the resident dashboard choose **Report Flooding**.
2. Pick a severity, describe the situation, allow location access if prompted, and submit.

Expected: the modal closes with a confirmation message.

☐ Pass ☐ Fail Notes: _____

T13.2 Staff verify the report**STAFF**

1. Open **Flood Reports**.
2. Find the pending report, mark it **Reviewed**, then **Resolved**.

Expected: the report shows the reporter, barangay, severity and description, with a map link if coordinates were attached. The status tabs and counts update at each step, and every change appears in the Audit Log.

☐ Pass ☐ Fail Notes: _____

14 Automated test suite

Manual testing shows the system works today; the automated suite shows it keeps working.

T14.1 Run the full suite

TERMINAL

1. Run:

```
docker compose exec app php artisan test
```

Expected: 259 passing tests and no failures, in well under a minute. The suite runs against an isolated in-memory database and never touches your demonstration data.

☐ Pass ☐ Fail Notes: _____

WORTH POINTING OUT TO AN ADVISER

Several tests exist specifically to prove the system refuses to do something: a forged reading creates no alert and sends no notification; an auto-mode device rejects manual control; a simulated message is never recorded as delivered; a resident cannot read another resident's notifications. Tests that assert a refusal are usually the ones worth showing.

15 Coverage summary

Map each manuscript module to the tests that demonstrate it. Complete this column as you go.

#	Module (Table 17)	Tests	Result
1	Account Management	T1.1 – T1.9	☐ Pass ☐ Fail
2	Dashboard & Visualisation	T2.1 – T2.4	☐ Pass ☐ Fail
3	Notification & Alerts (UI)	T3.1 – T3.2	☐ Pass ☐ Fail
4	IoT Telemetry & Device Management	T4.1 – T4.8	☐ Pass ☐ Fail
5	Smart Actuation	T5.1 – T5.6	☐ Pass ☐ Fail
6	Flood Prediction	T6.1 – T6.2	☐ Pass ☐ Fail
7	Automated Alert Dispatch	T7.1 – T7.6	☐ Pass ☐ Fail
8	Reports & Historical Analytics	T8.1 – T8.4	☐ Pass ☐ Fail
9	User & Access Management	T9.1 – T9.5	☐ Pass ☐ Fail
10	System Configuration	T10.1 – T10.3	☐ Pass ☐ Fail
11	Evacuation Center Management	T11.1 – T11.5	☐ Pass ☐ Fail
12	Audit & Compliance	T12.1 – T12.5	☐ Pass ☐ Fail
★	Community Flood Reports	T13.1 – T13.2	☐ Pass ☐ Fail
★	Automated regression suite	T14.1	☐ Pass ☐ Fail

Known limitations to declare, not hide

If an adviser asks about any of the following, these are the correct answers. Each is a stated design decision, not an oversight.

Area	Position
SMS delivery	Simulated by default and recorded as such. A live Philippine gateway can be enabled by configuration alone.
Push notifications	Not implemented in this version.
Prediction model	Ordinary least-squares linear regression. Confidence is goodness of fit, not forecast probability. LSTM is future work once a multi-season dataset exists.

Actuation	Fully implemented in software; demonstrated with low-voltage prototype hardware. Does not control municipal infrastructure.
Evacuation routing	A direct heading to the nearest available shelter. No road-routing engine is integrated.
Weather data	Simulated unless an OpenWeatherMap key is configured.
Field hardware	Sensor nodes are on order. The ingestion path they will use is already live and is proven by T4.2 – T4.6.

Walkthrough sign-off

Tests completed: ____ of 67 · Passed: ____ · Failed: ____

Student / Proponent · Date

Adviser · Date

Remarks:
